

Algebra: Solving Log Equations (Properties)

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DIRECTIONS

Apply Product, Quotient, or Power Rules to solve each logarithmic equation for x.

- 1 Solve using the Quotient Rule:

$$\log_{10} 32 - \log_{10} 4 = \log_{10} x$$

Answer: _____

- 2 Apply the Power Rule and simplify:

$$\log_3 4^2$$

Answer: _____

- 3 Apply the Power Rule and simplify:

$$\log_5 5^2$$

Answer: _____

- 4 Solve using the Quotient Rule:

$$\log_2 24 - \log_2 3 = \log_2 x$$

Answer: _____

- 5 Apply the Power Rule and simplify:

$$\log_5 5^2$$

Answer: _____

- 6 Apply the Power Rule and simplify:

$$\log_2 2^3$$

Answer: _____

- 7 Solve using the Quotient Rule:

$$\log_2 32 - \log_2 4 = \log_2 x$$

Answer: _____

- 8 Solve using the Product Rule:

$$\log_2 x + \log_2 4 = 3$$

Answer: _____

- 9 Solve using the Quotient Rule:

$$\log_2 6 - \log_2 1 = \log_2 x$$

Answer: _____



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Answer Key & Solutions

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TEACHER NOTES Product Rule: $\log(AB)=\log A+\log B$. Quotient Rule: $\log(A/B)=\log A-\log B$. Check domain!

1 Solve using the Quotient Rule:

$$= \quad \quad \quad x = 8$$
$$\log_{10} 32 - \log_{10} 4 = \log_{10} x$$

2 Apply the Power Rule and simplify:

$$= \quad \quad \quad 2 \cdot \log_3 4$$
$$\log_3 4^2$$

3 Apply the Power Rule and simplify:

$$= \quad \quad \quad 2 \cdot \log_5 5$$
$$\log_5 5^2$$

4 Solve using the Quotient Rule:

$$= \quad \quad \quad x = 8$$
$$\log_2 24 - \log_2 3 = \log_2 x$$

5 Apply the Power Rule and simplify:

$$= \quad \quad \quad 2 \cdot \log_5 5$$
$$\log_5 5^2$$

6 Apply the Power Rule and simplify:

$$= \quad \quad \quad 3 \cdot \log_2 2$$
$$\log_2 2^3$$

7 Solve using the Quotient Rule:

$$= \quad \quad \quad x = 8$$
$$\log_2 32 - \log_2 4 = \log_2 x$$

8 Solve using the Product Rule:

$$= \quad \quad \quad x = 2$$
$$\log_2 x + \log_2 4 = 3$$

9 Solve using the Quotient Rule:

$$= \quad \quad \quad x = 6$$
$$\log_2 6 - \log_2 1 = \log_2 x$$

